

KOYNAS, Russian Federation

WMO: 225830

Lat: 64.750N

Lon: 47.650E

Elev: 64

StdP: 100.56

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.5	(c) -33.9	(d) -40.2	(e) 0.1	(f) -37.3	(g) -36.7	(h) 0.1	(i) -33.9	(j) 7.2	(k) -9.1	(l) 6.3	(m) -7.6	(n) 0.5	(o) 320	(p) 0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	28.3	18.4	25.9	17.3	23.6	16.3	19.6	25.5	18.5	24.0	17.2	22.3	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	22.0	16.2	11.6	20.9	15.0	10.7	19.5	56.2	25.5	52.6	24.2	48.6	22.4	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 6.1	(b) 5.2	(c) 4.4	(e) -39.7	(f) 31.1	(g) 4.0	(h) 1.7	(i) -42.6	(j) 32.4	(k) -44.9	(l) 33.4	(m) -47.2	(n) 34.4	(o) -50.1	(p) 35.7		
			DB													
			WB													
				-39.7	21.6	3.8	1.4	-42.4	22.6	-44.6	23.4	-46.7	24.2	-49.4	25.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-15.3	-15.1	-7.7	-0.4	6.0	11.8	16.1	12.5	7.7	1.4	-7.2	-11.5		
	DBStd	12.44	9.50	9.64	6.58	5.30	5.51	4.99	4.53	3.66	3.68	3.97	7.58	9.07		
	HDD10.0	4097	783	702	550	313	152	36	6	15	88	268	517	666		
	HDD18.3	6759	1042	935	808	563	383	205	99	186	320	527	767	924		
	CDD10.0	421	0	0	0	1	27	90	193	91	18	1	0	0		
	CDD18.3	42	0	0	0	0	1	9	28	4	0	0	0	0		
	CDH23.3	511	0	0	0	0	28	123	320	39	1	0	0	0		
	CDH26.7	127	0	0	0	0	7	32	81	7	0	0	0	0		
(14) Wind	WSAvg	1.9	1.7	1.7	2.0	2.1	2.2	2.1	1.9	1.8	2.0	2.1	1.8	1.8		
(15) Precipitation	PrecAvg	609	36	30	30	39	50	65	68	68	64	62	52	45		
	PrecMax	739	77	57	76	77	98	162	136	110	107	113	90	79		
	PrecMin	479	16	7	6	12	18	10	13	40	22	16	12	10		
	PrecStd	69	14	13	16	17	20	34	31	22	24	23	18	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.7	6.8	17.3	27.3	30.0	31.6	27.2	21.2	13.0	5.9	2.5		
		MCWB	0.4	1.1	2.5	10.1	16.2	18.8	20.3	18.5	15.2	10.7	5.0	1.8		
	2%	DB	-0.5	0.5	3.5	12.3	22.6	27.3	29.0	23.8	17.5	10.2	3.7	1.1		
		MCWB	-0.8	-0.1	0.8	6.6	13.3	17.6	19.1	17.2	13.3	8.5	3.1	0.6		
	5%	DB	-1.8	-0.8	2.0	9.1	19.2	24.3	26.9	21.4	15.1	8.4	2.2	0.3		
		MCWB	-2.2	-1.4	0.2	4.5	11.6	15.9	18.0	16.0	12.2	7.3	1.7	-0.2		
	10%	DB	-3.3	-2.4	1.0	6.8	16.1	21.2	24.9	19.2	13.4	6.8	1.0	-0.8		
		MCWB	-3.8	-3.0	-0.3	3.2	10.3	14.5	17.3	15.0	11.2	6.0	0.6	-1.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	1.2	3.4	10.2	16.9	20.0	21.8	19.9	15.8	10.9	5.2	1.9		
		MCDB	0.5	1.6	6.1	17.1	24.8	27.4	28.9	25.3	19.4	12.7	5.7	2.3		
	2%	WB	-0.8	0.1	1.5	7.0	14.4	18.5	20.3	18.2	14.1	8.7	3.2	0.7		
		MCDB	-0.6	0.5	2.9	11.5	20.8	25.4	26.3	21.7	16.7	9.8	3.7	1.0		
	5%	WB	-2.1	-1.3	0.7	5.1	12.3	16.9	19.1	16.6	12.7	7.3	1.8	-0.1		
		MCDB	-1.8	-0.8	1.7	8.3	18.1	22.5	24.9	20.1	14.7	8.2	2.1	0.3		
	10%	WB	-3.7	-2.9	-0.4	3.6	10.5	15.3	18.1	15.5	11.3	6.1	0.7	-1.2		
		MCDB	-3.3	-2.4	0.9	6.1	15.6	20.5	23.4	18.6	13.1	6.8	1.0	-0.7		
(35) Mean Daily Temperature Range	MDBR		6.9	8.3	10.0	9.3	10.0	9.9	10.3	8.4	6.7	3.9	4.9	6.1		
	5% DB	MCDBR	5.2	4.9	8.3	13.2	16.2	14.8	14.5	12.9	10.5	5.8	3.6	4.4		
		MCWBR	5.2	4.7	6.5	8.2	8.3	6.8	6.2	6.8	6.7	4.7	3.6	4.3		
	5% WB	MCDBR	4.9	4.8	6.7	11.4	14.5	12.8	12.7	10.6	9.2	5.7	3.6	4.3		
		MCWBR	5.0	4.7	5.6	7.6	7.9	6.6	6.1	6.3	6.3	4.9	3.7	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.216	0.276	0.307	0.342	0.343	0.348	0.395	0.359	0.329	0.297	N/A	0.214		
	taud		1.919	2.058	2.109	2.131	2.401	2.489	2.358	2.439	2.510	2.339	N/A	1.933		
	Ebn at Noon		289	597	756	818	865	869	806	806	759	597	N/A	81		
	Edh at Noon		25	63	94	118	100	94	104	88	66	49	N/A	9		
(44) All-Sky Solar Radiation	RadAvg		0.14	0.75	2.16	3.83	4.90	5.22	5.08	3.40	1.78	0.66	0.17	0.04		
	RadStd		0.02	0.12	0.27	0.38	0.39	0.64	0.46	0.32	0.18	0.09	0.03	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page